

PYMBLE LADIES' COLLEGE
MATHEMATICS EXTENSION 1
TRIAL HSC EXAMINATION

2005

Reading Time: 5 minutes
Working Time: 2 hours

Instructions to students:

- Write using blue or black biro.
- All questions may be attempted.
- Diagrams are not to scale.
- All necessary working should be shown in every question.
- Your name and your teacher's name may be written before you begin the assessment.
- Start each question in a new booklet.
- Marks may be deducted for careless or untidy work.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

Question 1 (12 marks) Use a separate writing booklet

MARKS

(a) Differentiate $\tan^{-1}(2x)$.

2

(b) Solve $\frac{3}{2-x} < 6$.

3

(c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2x}$

2

(d) Find the coordinates of the point $P(x, y)$ which divides the interval joining the points $A(-6, 2)$ and $B(4, 7)$ externally in the ratio 3: 2.

2

(e) Evaluate $\int_0^{\sqrt{3}} \frac{x}{\sqrt{1+x^2}} dx$, using the substitution $u = 1+x^2$.

3

Question 2 (12 marks) Use a separate writing booklet

MARKS

- (a) (i) Show that $(x-3)$ is a factor of $x^3 + 3x^2 - 9x - 27$. 1
- (ii) Hence, or otherwise, factorise $x^3 + 3x^2 - 9x - 27$ completely. 2
- (b) Find the acute angle between the lines $x - 2y + 1 = 0$ and $x + 3y + 2 = 0$. 2
- (c) (i) Express $\sqrt{3} \sin x - \cos x$ in the form $A \sin(x - \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- (ii) Hence, or otherwise, solve $\sqrt{3} \sin x - \cos x = 1$ in the domain $0 \leq x \leq 2\pi$. 2
- (d) Find the term independent of x in the expansion of $\left(3x - \frac{1}{2x^2}\right)^9$. 3

Question 3 (12 marks) Use a separate writing booklet

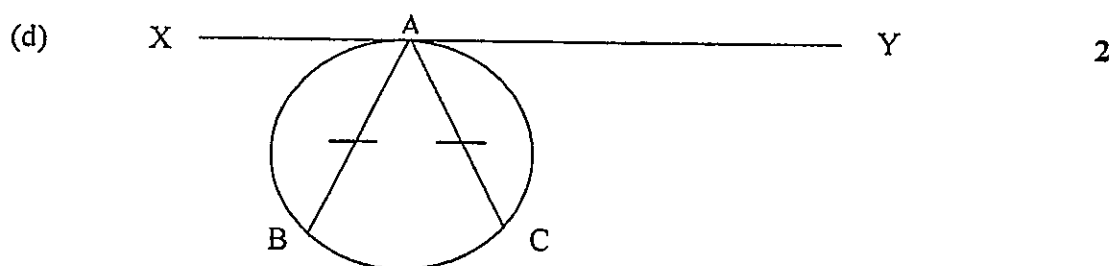
MARKS

(a) Evaluate $\sin^{-1}\left(\cos\frac{2\pi}{3}\right)$. 1

(b) State the domain and range of $y = 2\sin^{-1} 3x$. 2

(c) (i) Differentiate with respect to x : $x\sin^{-1} x + \sqrt{1-x^2}$. 2

(ii) Hence evaluate $\int_0^1 \sin^{-1} x \, dx$. 2



Given that $AB = AC$ and XY is a tangent to circle ABC at A , prove that XY is parallel to BC .

(e) Use mathematical induction to prove that for all positive integers n 3

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

Question 4 (12 marks) Use a separate writing booklet

MARKS

Consider the function $f(x) = \frac{e^x}{e^x + 2}$.

- | | | |
|-------|--|---|
| (i) | Show that $f(x)$ has no stationary points. | 3 |
| (ii) | Find the coordinates of the point of inflexion given that
$f''(x) = \frac{2e^x (2 - e^x)}{(e^x + 2)^3}$ | 2 |
| (iii) | Show that $0 < f(x) < 1$ for all x . | 2 |
| (iv) | Sketch the curve $f(x)$. | 2 |
| (v) | Explain why $f(x)$ has an inverse function. | 1 |
| (vi) | Find the inverse function $y = f^{-1}(x)$. | 2 |

Question 5 (12 marks) Use a separate writing booklet

MARKS

- (a) The points P ($4p, 2p^2$) and Q ($4q, 2q^2$) lie on the parabola $x^2 = 8y$.
- | | | |
|-------|--|---|
| (i) | Show that the equation of the tangent to the parabola at P is $y = px - 2p^2$. | 2 |
| (ii) | The tangent at P and the line passing through Q parallel to the y axis intersect at T.
Show that the coordinates of T are ($4q, 4pq - 2p^2$). | 1 |
| (iii) | Find coordinates of M, the midpoint of PT. | 2 |
| (iv) | Find the cartesian equation of the locus of M when $pq = 1$. | 1 |
- (b) The acceleration of a particle P moving in a straight line is given by $\frac{d^2 x}{dt^2} = 3x(x - 4)$, where x metres is the displacement of the particle from the origin O and t is the time in seconds. Initially the particle is at O and its velocity is $4\sqrt{2}$ m/s.
- | | | |
|-------|--|---|
| (i) | Show that $v^2 = 2(x^3 - 6x^2 + 16)$ where v is the velocity of P. | 3 |
| (ii) | Calculate the velocity and acceleration of P at $x = 2$. | 2 |
| (iii) | In which direction does P move from $x = 2$? Give a reason for your answer. | 1 |

Question 6 (12 marks) Use a separate writing booklet

MARKS

- (a) Using the fact that $(1+x)^3(1+x)^8=(1+x)^{11}$, show that
 ${}^3C_0 {}^8C_3 + {}^3C_1 {}^8C_2 + {}^3C_2 {}^8C_1 + {}^3C_3 {}^8C_0 = {}^{11}C_3$

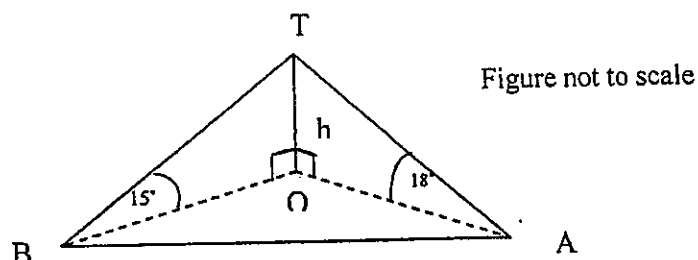
2

- (b) A sphere of ice cream sits on top of a cone. As the ice cream melts, maintaining its spherical shape, the radius reduces at a constant rate of 0.25 centimetres per minute. [The volume of a sphere is given by the formula $V = \frac{4}{3}\pi r^3$.]

Find the rate of change of the volume of the ice cream with respect to time when the radius is 4cm.

2

- (c)



3

The angle of elevation of a hill OT, at the point A due south of the hill, is 18° . From B, due west of A, the angle of elevation is 15° . If AB is 6 km, find the height, h of the hill. Give your answer to the nearest 10 metres.

- (d) The displacement x (in metres) of a particle is given by $x = 5\cos(4\pi t)$, where t is in seconds.

- (i) Show that the acceleration of the particle can be expressed in the form $\ddot{x} = -n^2x$.

2

- (ii) Determine the maximum speed of the particle and when it first occurs.

2

- (iii) Express v^2 in terms of x , where v is the velocity of the particle.

1

Question 7 (12 marks) Use a separate writing booklet

MARKS

- (a) A particle P is projected from a point on horizontal ground with velocity V at an angle of projection α . You may assume that the equations of motion are:

$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = V \sin \alpha - gt$$

$$x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2$$

- (i) Show that the particle's maximum height is $\frac{V^2 \sin^2 \alpha}{2g}$. 2

- (ii) A second particle Q is projected from the same point on horizontal ground with velocity $V \times \sqrt{\frac{5}{2}}$ at an angle $\frac{\alpha}{2}$ to the horizontal. If the maximum height is the same for both particles, show that $\alpha = \cos^{-1} \frac{1}{4}$. 4

- (b) (i) Prove that the graph of $y = \ln x$ is concave down for all $x > 0$. 2

- (ii) Sketch the graph of $y = \ln x$. 2
Suppose $0 < a < b$ and consider the points A $(a, \ln a)$ and B $(b, \ln b)$ on the graph of $y = \ln x$. Find the coordinates of the point P that divides the line segment AB in the ratio 2:1.

- (iii) By using part (ii) deduce that $\frac{1}{3} \ln a + \frac{2}{3} \ln b < \ln \left(\frac{1}{3}a + \frac{2}{3}b \right)$ 2

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

QUESTION 1

a. $\frac{d}{dx} \tan^{-1}(2x)$

$$= \frac{1}{1 + (2x)^2} \cdot 2$$

$$= \frac{2}{1 + 4x^2}$$

b. $\frac{3}{2-x} < 6$

METHOD 1: Multiply by the square of the Denominator

$$\frac{3}{2-x} \cdot (2-x)^2 < 6(2-x)^2$$

$$3(2-x) < 6(4-4x+x^2)$$

$$2-x < 2(4-4x+x^2)$$

$$2-x < 8-8x+2x^2$$

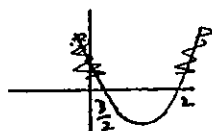
$$0 < 2x^2 - 7x + 6$$

$$2x^2 - 7x + 6 > 0$$

$$(2x-3)(x-2) > 0$$

Let $(2x-3)(x-2) = y$

WANT $y > 0$



Ans. $x < \frac{1}{2}$ or $x > 2$

METHOD 2: Consider denominator POSITIVE and NEGATIVE

Consider $2-x > 0$ or $2-x < 0$

i.e. $x < 2$

$x > 2$

Q1

then $3 < 6(2-x)$ or $3 > 6(2-x)$

$$3 < 12-6x \quad x > \frac{1}{2}$$

$$6x < 9$$

$$x < \frac{1}{2}$$

and since $x > 2$

and Since $x < 2$

then

then

$$\underline{x < \frac{1}{2}}$$

or

$$\underline{x > 2}$$

METHOD 3: Using "CRITICAL" POINTS

c. $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2x}$

$$= \frac{1}{2} \cdot \frac{1}{5} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{\frac{x}{5}}$$

$$= \frac{1}{10} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{\frac{x}{5}}$$

$$= \frac{1}{10}$$

N.B. As $x \rightarrow 0$ then

$$\frac{x}{5} \rightarrow 0$$

d. $A(-6, 2) \quad B(4, 7) \quad 3: -2$
 $x_1, y_1 \quad x_2, y_2 \quad k: L$

$$x = \frac{Lx_1 + kx_2}{k+L}$$

$$y = \frac{Ly_1 + ky_2}{k+L}$$

$$= \frac{-2x-6+3x4}{3-2}$$

$$= \frac{-2x2+3x7}{3-2}$$

$$= 24$$

$$= 17$$

$\therefore$ Required Point $P(x, y)$ is $(24, 17)$

Q3

d. $\angle XAB = \angle ACB$ Angle between the chord and the tangent is equal to the angle at the circumference in the alternate segment.

$\angle ACB = \angle ABC$ Base angles of isosceles triangle ABC

$\therefore$

Angles opposite equal sides of $\triangle ABC$

$\therefore \angle XAB = \angle ABC$ which represent alternate angles

As the alternate angles are equal then $XY \parallel BC$

e. To prove $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

STEP 1: To prove it true for $n=1$

$$\begin{aligned} \text{LHS} &= 1^2 \\ &= 1 \\ \text{RHS} &= \frac{1}{3} \times 1 \times (2 \times 1 - 1)(2 \times 1 + 1) \\ &= 1 \\ &= \text{LHS} \end{aligned}$$

$\therefore$ True for $n=1$

STEP 2: Assume it true for $n=k$

$$\text{i.e. } 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

STEP 3: To prove it true for $n=k+1$

Q3

e. cont'd

$$\text{LHS} = \underbrace{1^2 + 3^2 + 5^2 + \dots + (2k-1)^2}_{\text{from step 2}} + [2(k+1)-1]^2$$

$$= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2$$

$$= \frac{1}{3}(2k+1)[k(2k-1) + 3(2k+1)]$$

$$= \frac{1}{3}(2k+1)(2k^2 + 5k + 3)$$

$$= \frac{1}{3}(2k+1)(2k+3)(k+1)$$

$$\begin{aligned} \text{RHS} &= \frac{1}{3}(k+1)[2(k+1)-1][2(k+1)+1] \\ &= \frac{1}{3}(k+1)(2k+1)(2k+3) \\ &= \text{LHS} \end{aligned}$$

$\therefore$ True for $n=k+1$

STEP 4: Since it is true for $n=1$ and proven true for $n=1+1$ i.e. $n=2$ and for $n=2+1$ i.e. $n=3$ and so on, it is true for all positive integer values of n .

Question 4

$$\begin{aligned} \text{i. } f'(x) &= \frac{e^x(e^x+2) - e^x \cdot e^x}{(e^x+2)^2} \\ &= \frac{e^{2x} + 2e^x - e^{2x}}{(e^x+2)^2} \\ &= \frac{2e^x}{(e^x+2)^2} \end{aligned}$$

Q4

i. conc'd $f'(x) \neq 0$ as $2e^x \neq 0$ since $e^x > 0$ for all x

ii.
$$f''(x) = \frac{2e^x(2-e^x)}{(e^x+2)^3}$$

Points of inflexion occur when $f''(x) = 0$

i.e. $2e^x(2-e^x) = 0$

Either $e^x = 0$ or $2-e^x = 0$

No sol'n

$e^x = 2$

as $e^x > 0$

$x = \ln 2$

$$f''(0) = \frac{2e^0(2-e^0)}{(e^0+2)^3}$$

 $\uparrow$
 > 0

$$f''(\ln 2) = \frac{2e^{\ln 2}(2-e^{\ln 2})}{(e^{\ln 2}+2)^3}$$

 $= \frac{3 \times -1}{5^3}$

< 0

As there is a change in concavity then $x = \ln 2$ must be a point of inflexion.

$$f(\ln 2) = \frac{e^{\ln 2}}{e^{\ln 2} + 2}$$

 $= \frac{2}{2+2}$
 $= \frac{1}{2}$

Q4

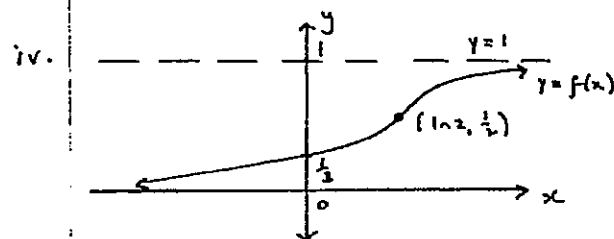
iii.
$$f(x) = \frac{e^x}{e^x+2}$$

As $e^x > 0$ then $\frac{e^x}{e^x+2} > 0$
 for all x

As $x \rightarrow \infty$, then $\frac{e^x}{e^x+2} \rightarrow 1$

As $x \rightarrow -\infty$, then $\frac{e^x}{e^x+2} \rightarrow 0$

$\therefore 0 < f(x) < 1$ for all x



$$f(0) = \frac{e^0}{e^0+2} = \frac{1}{3}$$

v. For every y -value there is only 1 x -value

OR ANY HORIZONTAL LINE CAN BE DRAWN
 AND IT CUTS $f(x)$ AT ONLY 1 POINT.

OR INCREASING FUNCTION OVER ENTIRE
 DOMAIN

Q4

vi.

$$y = \frac{e^x}{e^x + 2}$$

Inverse function is given by

$$x = \frac{e^y}{e^y + 2}$$

$$xe^y + 2x = e^y$$

$$e^y(1-x) = 2x$$

$$e^y = \frac{2x}{1-x}$$

$$y = \ln\left(\frac{2x}{1-x}\right)$$

QUESTION 5

a. i.

$$x^2 = 8y$$

$$y = \frac{x^2}{8}$$

$$\frac{dy}{dx} = \frac{x}{4}$$

$$\text{when } x = 4p, \quad \frac{dy}{dx} = \frac{4p}{4} = p$$

Eqn of tangent at $P(4p, 2p^2)$ is given by

$$y - y_1 = m(x - x_1)$$

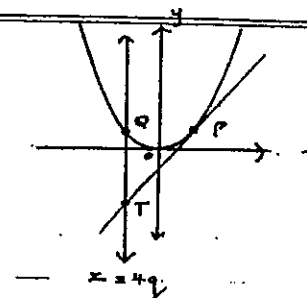
$$y - 2p^2 = p(x - 4p)$$

$$y - 2p^2 = px - 4p^2$$

$$\therefore y = px - 2p^2 \quad \text{--- (1)}$$

Q5

a. ii.



As parallel to the y-axis then again of the line must be $x = 4q$. (2)

As T lies on this line then x-value of T must be $4q$.

Sub (2) into (1)

$$y = p(4q) - 2p^2$$

$$y = 4pq - 2p^2$$

$\therefore$ Coords of T is $(4q, 4pq - 2p^2)$

iii.

$$P(4p, 2p^2)$$

$$x_1, y_1$$

$$T(4q, 4pq - 2p^2)$$

$$x_2, y_2$$

Midpoint M is given by

$$x = \frac{x_1 + x_2}{2}$$

$$y = \frac{y_1 + y_2}{2}$$

$$= \frac{4p + 4q}{2}$$

$$= \frac{2p^2 + 4pq - 2p^2}{2}$$

$$= \frac{4(p+q)}{2}$$

$$= \frac{4pq}{2}$$

$$= 2(p+q)$$

$$= 2pq$$

$\therefore M$ is $(2(p+q), 2pq)$

Q5

a. iv. $x = 2(p+q)$ $y = 2pq$
 Sub $pq=1$
 $x = 2(p+q)$ $y = 2$

As y is independent of p and q then this represents the locus of M i.e. the horizontal line $y=2$.

b. i.

$$a = 3x(x-4)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 3x^2 - 12x$$

$$\frac{1}{2} v^2 = \frac{3x^3}{3} - \frac{12x^2}{2} + C$$

$$\frac{1}{2} v^2 = x^3 - 6x^2 + C$$

Sub $x=0$, $v = 4\sqrt{2}$

$$\frac{1}{2} [16 \times 2] = C$$

$$\therefore C = 16$$

$$\frac{1}{2} v^2 = x^3 - 6x^2 + 16$$

$$v^2 = 2(x^3 - 6x^2 + 16)$$

ii. when $x=2$, $v^2 = 2(2^3 - 6 \times 2^2 + 16)$

$$v^2 = 0$$

$$\therefore v = 0 \text{ At rest}$$

$$a = 3 \times 2(2-4)$$

$$= -12 \text{ m s}^{-2}$$

Q5

b. iii. Moves to the left (towards origin) as $a < 0$ and $v = 0$.

The force will now push it backwards as the particle stopped momentarily at $x=2$.

Question 6

a. $(1+x)^2(1+x)^8 = (1+x)^{10}$

$$\left[\sum_{r=0}^2 {}^3C_r x^r \right] \left[\sum_{r=0}^8 {}^8C_r x^r \right] = \sum_{r=0}^{10} {}^{10}C_r x^r$$

To prove ${}^3C_0 {}^8C_2 + {}^3C_1 {}^8C_1 + {}^3C_2 {}^8C_0 = {}^{10}C_3$

RHS = ${}^{10}C_3$ represents the coefficient of x^3 term.

Q6

a. cont'd

LHS: Obtain x^3 term by

$${}^3C_0 {}^8P_3 x^3 + {}^3C_1 x {}^8P_2 x^2 + {}^3C_2 x^2 {}^8P_1 x + {}^3C_3 x^3 {}^8P_0$$

$$= x^3 [{}^3C_0 {}^8P_3 + {}^3C_1 {}^8P_2 + {}^3C_2 {}^8P_1 + {}^3C_3 {}^8P_0]$$

Hence coefficient of x^3 term is

$${}^3C_0 {}^8P_3 + {}^3C_1 {}^8P_2 + {}^3C_2 {}^8P_1 + {}^3C_3 {}^8P_0$$

Hence equating coefficients of x^3 term

$${}^3C_0 {}^8P_3 + {}^3C_1 {}^8P_2 + {}^3C_2 {}^8P_1 + {}^3C_3 {}^8P_0 = {}^4C_3$$

b.

$$\frac{dr}{dt} = -0.25$$

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\text{Want } \frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot -0.25$$

$$= -\pi r^2$$

$$\text{Sub } r = 4$$

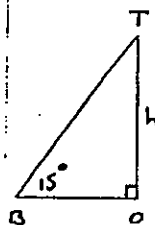
$$\frac{dV}{dt} = -4^2 \cdot \pi$$

$$= -16\pi \text{ cm}^3/\text{minute}$$

The volume is decreasing at the rate of $16\pi \text{ cm}^3/\text{minute}$.

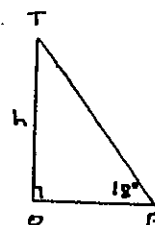
Q6

c.

In $\triangle BOT$ 

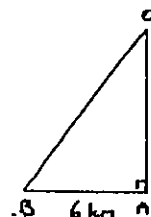
$$\tan 15^\circ = \frac{h}{OB}$$

$$OB = \frac{h}{\tan 15^\circ}$$

In $\triangle AOT$ 

$$\tan 18^\circ = \frac{h}{OA}$$

$$OA = \frac{h}{\tan 18^\circ}$$

In $\triangle AOB$ 

$$OA^2 + OB^2 = AB^2$$

Pythagoras' Thm

$$\frac{OA}{OB} = \tan 75^\circ = \frac{OA}{h}$$

$$OB = h \tan 75^\circ$$

$$\tan 72^\circ = \frac{OA}{h}$$

$$OA = h \tan 72^\circ$$

From $\triangle AOB$

$$OB^2 - OA^2 = 36 \quad \text{Pythagoras' Thm}$$

$$h^2 [\tan^2 75^\circ - \tan^2 72^\circ] = 36$$

$$h^2 = \frac{36}{\tan^2 75^\circ - \tan^2 72^\circ}$$

$$h = \frac{6}{\sqrt{\tan^2 75^\circ - \tan^2 72^\circ}}$$

Height of hill $\div 2.84 \text{ km}$
to nearest 10m.

$$\text{d.i. } x = 5 \cos(4\pi t)$$

$$\dot{x} = -5.5 \sin(4\pi t) \cdot 4\pi$$

$$\dot{x} = -20\pi \sin(4\pi t)$$

$$\ddot{x} = -20\pi \cdot 4\pi \cos(4\pi t)$$

$$\ddot{x} = -16\pi^2 \cdot 5 \cos(4\pi t)$$

$$\therefore \ddot{x} = -(4\pi)^2 x$$

This is of the form $\ddot{x} = -n^2 x$ where $n = 4\pi$

- d. ii. From part (i) $\ddot{x} = -20\pi \sin(4\pi t)$
 Max Speed occurs when $\sin(4\pi t) = 1$
 i.e. $|\ddot{x}| = 20\pi \text{ m/s}$

or $v^2 = n^2(a^2 - x^2)$

Sub $n = 4\pi$, $a = 5$

$v^2 = 16\pi^2(25 - x^2)$

Max Speed occurs at the centre of motion

i.e. Sub $x = 0$

$v^2 = 16 \times 25 \times \pi^2$

Speed $= |v| = 4 \times 5 \times \pi$
 $= 20\pi \text{ m/s}$

or Max Speed occurs at the centre of motion i.e. at $x = 0$

$\therefore 5 \cos(4\pi t) = 0$

$\cos(4\pi t) = 0$

$4\pi t = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$

$t = \frac{1}{8} \text{ seconds}$

Sub into (i)

$\ddot{x} = -20\pi \sin(4\pi \cdot \frac{1}{8})$
 $= -20\pi \sin \frac{\pi}{2}$

$\therefore \text{Max Speed} = 20\pi \text{ m/s}$

Answer: Max Speed is $20\pi \text{ m/s}$
 and first occurs after $\frac{1}{8}$ seconds

iii. $v^2 = n^2(a^2 - x^2)$

Sub $n = 4\pi$, $a = 5$

$v^2 = 16\pi^2(25 - x^2)$

Question 7

- a. i. Maximum Height occurs when $\dot{y} = 0$

i.e. $V \sin \alpha - gt = 0$

$gt = V \sin \alpha$

$t = \frac{V}{g} \sin \alpha$

Sub into $y = Vt \sin \alpha - \frac{1}{2}gt^2$

$y = V \cdot \frac{V}{g} \sin \alpha \cdot \sin \alpha - \frac{1}{2}g \left[\frac{V}{g} \sin \alpha \right]^2$

$y = \frac{V^2}{g} \sin^2 \alpha - \frac{V^2}{2g} \sin^2 \alpha$

$\therefore \text{Max Height } y_p = \frac{V^2 \sin^2 \alpha}{2g}$

- ii. Similarly Max Height for particle Q is

$y_a = \frac{(V \times \frac{\sqrt{2}}{2})^2 \sin^2(\frac{\pi}{2})}{2g}$

$y_a = \frac{5V^2 \sin^2(\frac{\pi}{2})}{4g}$

Since Max Height the same for both particles

then $y_p = y_a$

i.e. $\frac{V^2 \sin^2 \alpha}{2g} = \frac{5V^2 \sin^2(\frac{\pi}{2})}{4g}$

$\sin^2 \alpha = \frac{5}{2} \sin^2(\frac{\pi}{2})$

Using $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$

Then $\sin^2 \alpha = \frac{5}{4}(1 - \cos \alpha)$

Q7

a. ii. const: $1 - \cos^2 \alpha = \frac{5}{4} (1 - \cos \alpha)$

$$4 - 4\cos^2 \alpha = 5 - 5\cos \alpha$$

$$4\cos^2 \alpha - 5\cos \alpha + 1 = 0$$

$$(4\cos \alpha - 1)(\cos \alpha - 1) = 0$$

Either $4\cos \alpha - 1 = 0$ or $\cos \alpha - 1 = 0$

$$4\cos \alpha = 1$$

$$\cos \alpha = 1$$

$$\cos \alpha = \frac{1}{4}$$

$$\alpha = 0, \pi$$

rejected as

Not Applicable in.

not projected up

$$\therefore \alpha = \cos^{-1}\left(\frac{1}{4}\right)$$

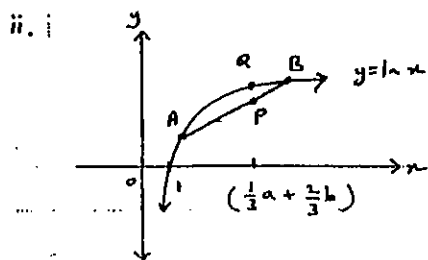
b. i. $y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x} = x^{-1}$$

$$\frac{d^2y}{dx^2} = -x^{-2} = -\frac{1}{x^2}$$

As $x^2 > 0$ for all x , then $-\frac{1}{x^2} < 0$

Hence $\frac{d^2y}{dx^2} < 0$ for all $x > 0$



Q7

b. ii. const: $A(a, \ln a) \quad B(b, \ln b) \quad 2:1$

$$x_1 \quad y_1 \quad x_2 \quad y_2 \quad k \quad L$$

Hence coordinates of P is given by:

$$x = \frac{x_1 L + x_2 k}{k+L}$$

$$y = \frac{y_1 L + y_2 k}{k+L}$$

$$x = \frac{a+2b}{3}$$

$$y = \frac{\ln a + 2\ln b}{3}$$

$$= \frac{1}{3}(a+2b)$$

$$= \frac{1}{3} \ln(ab^2)$$

$\therefore P$ is $\left(\frac{1}{3}a + \frac{2}{3}b, \frac{1}{3}\ln a + \frac{2}{3}\ln b\right)$

iii. Q (directly above P) is on the curve $y = \ln x$.

So for $y = \ln x$

$$\text{Sub } x = \frac{1}{3}a + \frac{2}{3}b$$

$$\text{then } y_Q = \ln\left(\frac{1}{3}a + \frac{2}{3}b\right)$$

As $0 < a < b$ and the curve is concave down for all $x > 0$, then the curve must lie above the line segment AB .

Hence the y -value of the point P must be below the y -value of the point Q on the curve.

i.e. $y_P < y_Q(\ln x)$

$$\frac{1}{3}\ln a + \frac{2}{3}\ln b < \ln\left(\frac{1}{3}a + \frac{2}{3}b\right)$$